UMGS: UAV Multispectral 3D Gaussian Splatting for Spectral-Index Synthesis – Supplementary Material

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1 Protocol Notes and Scope

This supplement aligns with the current main-paper draft and records additional tables that are useful for reproducibility and reviewer inspection.

The main paper focuses on a 10-scene UAV-focused evaluation suite: 7 raw UAV captures and 3 aligned aerial multispectral scenes. This supplement separately reports four additional official aligned scenes whose viewpoints are not part of the UAV-focused main scope. The retained-subset diagnostic for raw unaligned input is reported as an applicability check and should not be interpreted as the main raw full-scene comparison protocol.

Depth diagnostics follow the relative reference-depth protocol when the reference source passes validation. Since raw UAV reconstructions are not in a globally calibrated metric coordinate system, all thresholds are relative-depth ratios rather than metric distances. We exclude failed dense-reference or frame-misaligned attempts from quantitative tables.

Resolution-aligned comparison views. The dagger marker (†) denotes UMGS renders bilinearly resized to the external adapter resolution solely for common-mask comparison. These rows are comparison views, not distinct training variants, and are not used to define the UMGS method.

Grouped split sanitization. For official aligned scenes with inconsistent split and transform metadata, we sanitize splits at the capture-group level against the available transforms. A D/G/R/RE/NIR group is removed only when one or more required members are absent from the transform file, making it unusable as a complete multispectral comparison unit. This is a protocol repair for data completeness, not a performance-based selection.

Depth reference. Depth diagnostics are validation-gated reference-depth checks, not dense MVS or LiDAR ground truth. The current main evidence uses references that pass visual, support, and camera-frame checks. Attempts whose fused point clouds were empty, whose meshes collapsed, or whose rendered reference views were frame-misaligned are retained only as audit artifacts and are not averaged into the reported depth table. The geometry claims therefore concern relative reference-depth consistency under held-out probe views, not absolute dense geometry accuracy.

2 Scene Name Mapping

The main paper uses publication-facing scene names. Internal directory names are kept only for reproducibility scripts and data manifests.

Table 1. Publication-facing scene names and internal identifiers.

Published name	Internal ID	Source
UMGS-Field	`self_m3m`	Project-owned UAV M3M capture
Vineyard-01	`raw001`	Zenodo vineyard record
Vineyard-02	`raw002`	Zenodo vineyard record
Vineyard-03	`raw003`	Zenodo vineyard record
Vineyard-04	`raw004`	Zenodo vineyard record
Vineyard-05	`raw005`	Zenodo vineyard record
Vineyard-06	`raw006`	Zenodo vineyard record
Aerial-Golf	`ms_golf`	MS-Splatting dataset
Aerial-Lake	`ms_lake`	MS-Splatting dataset
Aerial-Solar	`ms_solar`	MS-Splatting dataset

3 Additional Non-UAV Official Scene-Wise External Comparison

Table 4 reports the four additional official aligned scenes that are intentionally kept out of the main UAV-focused tables. Table 5 provides their scene-wise common-mask comparison between UMGS-J† and MS-Splatting. We also include the full seven-scene official aligned average for reproducibility.

4 Depth-Reference Status and Visualization

We keep the relative-depth evaluator and validation scripts in the released code, and report quantitative depth only for validated references. The main diagnostic table uses four validated scenes: UMGS-Field, Vineyard-01, Vineyard-03, and Aerial-Golf. Other attempted references are retained as audit artifacts only. For these representative scenes, we provide 10-held-out visual grids over RGB/G/R/RE/NIR where the relevant depth adapters are available. These grids show that UMGS-I and UMGS-J produce smoother relative-error maps, while the retained MS-Splatting adapter has lower support and larger local deviations even after coordinate-frame correction.

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The retained raw diagnostic is intentionally separate from the full raw protocol. It evaluates the subset that the external pipeline can retain, not the complete raw UAV reconstruction path.

5 Why Unfrozen-Support Is Not a Product Model

The Unfrozen-support branch is included as a single-band upper-bound ablation. It starts from the same RGB anchor as UMGS, but allows band-specific geometry and opacity updates in Stage 2. This can improve selected per-band render metrics, but it breaks the shared-support contract required by model-level spectral products.

The failure is concrete in our audit logs. On Aerial-Lake, Unfrozen-support improves several band PSNR values, but product construction fails with an xyz mismatch between bands:

$$\max |\Delta xyz| = 6.84 \times 10^{-2}.$$

Scratch-MS training diverges further and triggers Gaussian-count mismatch across bands. These variants can still produce independent band images, and one could compute post-hoc 2D ratios after rendering; however, those ratios are not defined over a common 3D Gaussian support and therefore are not used as valid model-level multispectral products in our protocol.

6 UMGS-J Initialization Ablation (UMGS-J vs UMGS-JR)

The main paper keeps UMGS-JR out of the main tables and reports UMGS-J as the representative joint branch. This section records the initialization ablation that motivated that decision. Both variants use the same geometry-locked joint multispectral training design and export per-band evaluation views from a unified checkpoint. They differ only in appearance-bank initialization.

UMGS-J initializes each spectral bank from the reset-appearance state. UMGS-JR initializes each spectral bank from the RGB-tied carrier.

Table 7 compares these variants on the two joint-branch pilot scenes used during method screening. UMGS-JR does not improve the branch: it degrades image/index quality on both pilot scenes.

7 Takeaway

The supplementary tables support three practical points used by the paper: (1) additional official aligned scenes remain available for reproducibility but are outside the main UAV-focused scope, (2) depth-reference evidence is reported only when the reference source passes validation, and (3) UMGS-JR is an initialization ablation rather than a promoted branch.

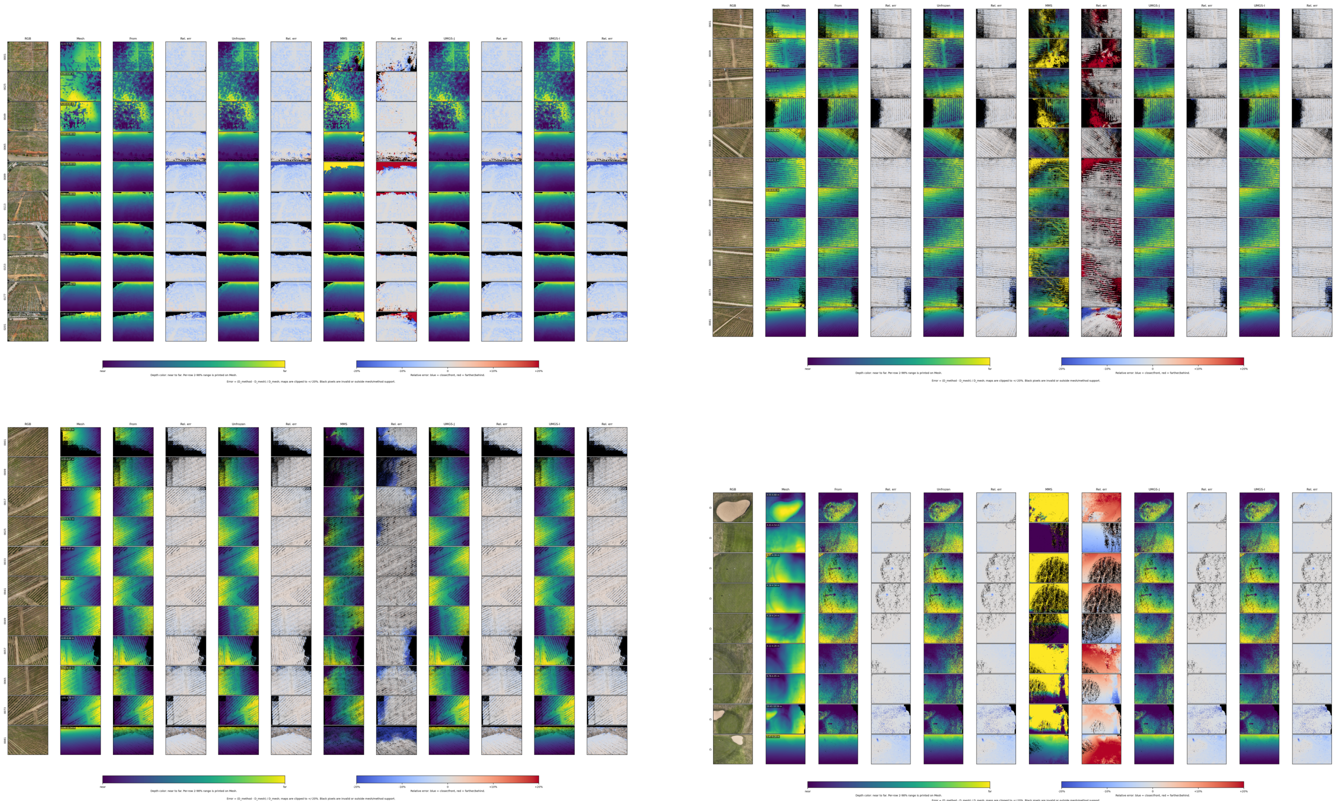

Fig. 1. RGB-band reference-depth visual grids for the four validated scenes used in the depth diagnostic: UMGS-Field, Vineyard-01, Vineyard-03, and Aerial-Golf. Each grid follows the main-paper layout: input image, mesh reference depth, and method depth/error pairs for Scratch-MS, Unfrozen-support, retained MS-Splatting, UMGS-J, and UMGS-I. Full RGB/G/R/RE/NIR grids are included in the released data bundle.

Table 2. Scene-level validation-gated reference-depth consistency. Values average G/R/RE/NIR for each validated scene. These diagnostics use scene-specific reference meshes and held-out probe cameras; they are consistency checks, not absolute geometry accuracy.

Scene	Method	Valid ↑	AbsRel ↓	Ag@1% ↑	Ag@5% ↑	Ag@10% ↑
UMGS-Field	UMGS-I/J	1.000	0.047	0.273	0.728	0.904
UMGS-Field	Scratch-MS	0.792	0.022	0.335	0.713	0.783
UMGS-Field	Unfrozen-support	1.000	0.048	0.264	0.722	0.904
UMGS-Field	MS-Splatting retained	1.000	0.465	0.485	0.736	0.808
Vineyard-01	UMGS-I/J	0.917	0.013	0.546	0.901	0.912
Vineyard-01	Scratch-MS	0.063	0.013	0.041	0.061	0.063
Vineyard-01	Unfrozen-support	0.928	0.013	0.547	0.911	0.922
Vineyard-01	MS-Splatting retained	0.644	0.178	0.383	0.480	0.494
Vineyard-03	UMGS-I/J	0.909	0.014	0.416	0.896	0.908
Vineyard-03	Scratch-MS	0.074	0.016	0.028	0.071	0.074
Vineyard-03	Unfrozen-support	0.921	0.014	0.419	0.907	0.920
Vineyard-03	MS-Splatting retained	0.733	0.048	0.459	0.598	0.628
Aerial-Golf	UMGS-I/J	0.979	0.017	0.624	0.941	0.966
Aerial-Golf	Scratch-MS	0.710	0.030	0.442	0.663	0.675
Aerial-Golf	Unfrozen-support	0.981	0.017	0.606	0.944	0.968
Aerial-Golf	MS-Splatting retained	0.826	0.089	0.065	0.290	0.510

Table 4. Additional official aligned non-UAV scenes, reported separately from the UAV-focused main-paper evaluation suite. RGB metrics for UMGS-J use the shared UMGS-I RGB anchor. Spectral metrics average G/R/RE/NIR.

Method	RGB PSNR	RGB SSIM	MS PSNR	MS SSIM	MS LPIPS	Idx. RMSE	Idx. SSIM
UMGS-I	21.67	0.690	25.75	0.763	0.262	0.108	0.710
UMGS-J	21.67	0.690	25.75	0.763	0.262	0.108	0.710

Table 5. Additional non-UAV official aligned scene-level common-mask comparison between UMGS-J† and MS-Splatting. PSNR/SSIM/LPIPS average G/R/RE/NIR; RMSE and SAM are spectral 4-band metrics.

Scene	UMGS-J PSNR	MMS PSNR	UMGS-J SSIM	MMS SSIM	UMGS-J LPIPS	MMS LPIPS	UMGS-J RMSE	MMS RMSE	UMGS-J SAM	MMS SAM
Aligned-Bud	24.77	25.09	0.797	0.781	0.225	0.257	0.0630	0.0601	6.46	7.34
Aligned-Fruit	24.18	23.81	0.698	0.701	0.318	0.306	0.0642	0.0670	8.12	8.66
Aligned-Garden	29.44	27.36	0.874	0.820	0.179	0.283	0.0378	0.0460	3.60	4.40
Aligned-Tree	26.45	24.83	0.812	0.757	0.251	0.325	0.0525	0.0644	5.90	7.07

Table 7. UMGS-J and UMGS-JR quality/cost comparison from pilot runs. Band metrics are G/R/RE/NIR means. Index metrics are NDVI/GNDVI/NDRE means.

Scene	Variant	Band SSIM ↑	Band PSNR ↑	Band LPIPS ↓	Index MAE ↓	Index RMSE ↓	Index PSNR ↑	Index SSIM ↑	Time (s) ↓
UMGS-Field	UMGS-J	0.7657	25.330	0.2113	0.0607	0.0863	27.701	0.7943	8129
UMGS-Field	UMGS-JR	0.7434	24.966	0.2061	0.0623	0.0880	27.519	0.7854	7339
Vineyard-01	UMGS-J	0.8804	28.690	0.1092	0.0339	0.0513	32.169	0.8948	5805
Vineyard-01	UMGS-JR	0.8590	27.847	0.1266	0.0375	0.0574	31.326	0.8800	6730